

Zahra Khotanlou

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SUMMARY

Machine learning developer with an MASc in Electrical and Computer Engineering (Pattern Analysis and Machine Intelligence) from the University of Waterloo with published and ongoing work spanning trustworthy AI, multi-agent systems, and deep learning for biological sequence data. Proficient across deep learning architectures (transformers, graph neural networks, diffusion models, object detection) and classical statistical ML, with hands-on experience training and evaluating models on GPU-accelerated, high-dimensional data pipelines. Comfortable moving quickly into newly forming problem areas, having applied deep learning across protein biology, LLM-based multi-agent systems, and computer vision. Strong foundation in core CS algorithms (Teaching Assistant for Data Structures & Algorithms) and fluent in AI-assisted development tooling (Claude Code, GitHub Copilot, Cursor) for high-velocity, scalable implementation. Experienced taking research from formulation through experimental validation to reproducible, tested code.

EDUCATION

University of Waterloo <i>MASc in Electrical & Computer Engineering</i>	Sept 2024 - Apr 2026 <i>Waterloo, ON</i>
- Core program: Pattern Analysis and Machine Intelligence - Teaching Assistant: ECE250 Data Structures & Algorithms (Winter 2026), ECE457 Intro to Machine Learning (Fall 2025) - Relevant coursework: Intro to AI (98/100), Multi-Agent Systems (95/100), NLP for ML (93/100), Deep Learning and Deployment (91/100)	
Shahid Beheshti University <i>B.Sc. in Computer Engineering</i>	Sept 2019 - Sept 2023 <i>Tehran, Iran</i>
Honors: Best Undergraduate Thesis (Dept. of Computer Engineering); merit-based scholarship for direct-entry graduate admission, awarded on academic standing	

SKILLS AND TOOLS

Machine Learning & NLP: PyTorch, TensorFlow, Scikit-learn, LightGBM, Hugging Face Transformers, Diffusion Models, Graph Neural Networks, Recommender Systems (Collaborative Filtering), vLLM, DeepSpeed, JAX
Generative AI: RAG pipelines, LLM orchestration, prompt engineering, structured outputs, Gemini API, OpenAI API
Programming: Python, Java, C++, TypeScript, C#, SQL
AI-Assisted Development: ChatGPT, Claude, Codex, Claude Code, GitHub Copilot, and Cursor for software design, implementation, debugging, testing, and documentation
Data & MLOps: Pandas, NumPy, ETL pipelines, relational databases, MLflow, DVC, Prometheus, Grafana
Cloud & Systems: AWS (Lambda, API Gateway, S3, IAM, CloudWatch — AWS Knowledge: Serverless, 2026), Kubernetes, Spark, Hadoop, Linux/Unix, computer networking, Docker, GPU-enabled workflows, distributed data pipelines

PROFESSIONAL EXPERIENCE

Research Assistant - Agentic AI for Value-Driven Scheduling <i>University of Waterloo, Supervisor: Prof. Edith Law</i>	June 2026 - Present
- Building a multi-agent LLM system for value elicitation and personalized scheduling, enabling users' values and preferences to guide AI-assisted decisions. - Designing an orchestration layer for request routing, agent coordination, and structured data exchange across multi-turn interactions. - Implementing persistent state and shared-memory components to maintain context across agent calls. - Developing explainability components to surface trade-offs in AI-assisted scheduling decisions.	
Research Assistant - Explainable & Fair AI <i>CHARM Lab, University of Waterloo, Supervisor: Prof. Amir-Hossein Karimi</i>	Sept 2024 - Apr 2026
- Multi-Agent Counterfactual Explanations (ACM FAccT 2026, github.com/zkhotanlou/Many-to-Many-AR): Developed a novel many-to-many algorithmic recourse framework that models interactions among users and decision providers under shared capacity constraints. - Formulated the problem as capacitated weighted bipartite matching and implemented the complete optimization, experimentation, and visualization pipeline in Python. - Attained 97-99% of the individual-welfare upper bound through capacity redistribution with minimal system modification, validated on real-world datasets (Credit, COMPAS). - RecourseBench (github.com/zkhotanlou/recourse_benchmarks): Built a scalable Python benchmarking platform integrating 22 heterogeneous ML methods and supporting 137 method, dataset, and model configurations. - Designed standardized interfaces that simplified the integration and maintenance of ML methods across multiple frameworks. - Developed automated CI, reproducibility, and regression tests to validate installation, execution reliability, output consistency, and fidelity to reported results.	
Research Assistant - Protein Property Prediction <i>National Institute of Genetic Engineering and Biotechnology (NIGEB)</i>	March 2023 - September 2024
- Built ToxTrans, a protein toxicity prediction module: extracted embeddings from protein sequences using protein language models (ProtT5), selected top features via ANOVA F-value, and classified using a tuned, GPU-accelerated LightGBM. - Trained and evaluated on 8,233 protein sequences from UniProt; achieved 93.76% accuracy, 0.98 AUC, and 0.87 MCC on independent testing, outperforming all prior methods including ToxMVA (+2.16% accuracy, +0.04 MCC). - Used GPU-enabled Python workflows for protein-model experimentation and compared sequence representation families to identify robust features for high-dimensional biological data.	
Software Engineer Intern <i>Star Academy</i>	June - October 2021
Built a Java-based full-text search engine using inverted-index architecture with Boolean (AND/OR/NOT) query support; refactored for	

- Clean Code/SOLID principles with full unit test coverage, then re-implemented in C# (TDD) and migrated to Elasticsearch (Nest) with Kibana visualization.
- Exposed the search service via a RESTful API in ASP.NET Core with Entity Framework Core and an Angular front end, with CI/CD pipelines for automated build and test execution.
- Designed and implemented an ETL (Extract, Transform, Load) data pipeline as a team capstone project: SQL Server/CSV ingestion, filter/join/aggregation transforms, YAML-based scenario persistence, and a web-based interface for managing pipeline runs.

Backend Developer

July 2020 - March 2021

Khishavere

- Built backend APIs and a web crawler to extract and structure student exam results into a database, processing raw HTML into cleaned, queryable records.
- Designed psychology-informed algorithms to analyze behavioral patterns in student performance across exam sessions.

SELECTED PROJECTS

Tim Hortons Cup Object Detection Pipeline

YOLOv8, Flask, FastAPI, MLflow, Prometheus, Grafana, DVC

- Built and deployed an end-to-end detection pipeline serving inference through Flask and FastAPI endpoints, with MLflow, DVC, Prometheus, and Grafana for tracking and monitoring.
- Conducted precision/recall-oriented failure analysis using CAM overlays, confusion matrices, scenario heatmaps, and UMAP, improving F1 from 0.942 to 0.959.
- Deployed a final model achieving 95.4% image-level accuracy and $F1 = 0.967$, monitored with real-time drift detection.

E-book Comments Sentiment Analysis ([GitHub](#))

HuggingFace Transformers, ParsBERT, Scikit-learn

- Fine-tuned the ParsBERT transformer on Persian-language customer reviews for binary sentiment classification, handling class imbalance through preprocessing and label-mapping strategies.

RAG-CFE: Retrieval-Augmented LLMs for Counterfactual Explanations ([GitHub](#))

Gemini 2.5 Flash, DiCE, RAG, LLMs, Prompt Engineering

- Designed an end-to-end RAG framework that combines DiCE-generated counterfactual candidates with LLM-based refinement to improve feasibility, plausibility, and actionability of recourse recommendations.
- Evaluated RAG-CFE against a DiCE baseline on validity, proximity, sparsity, feasibility, and plausibility, showing improved feasibility while maintaining prediction validity.

LSTM & GRU Stock Price Prediction ([GitHub](#))

PyTorch, LSTM, GRU, Time-Series Forecasting

- Implemented and compared LSTM and GRU recurrent architectures for forecasting stock closing prices over a 30-day window, evaluating forecasting accuracy across both architectures.

Traffic Sign Recognition Ensemble ([GitHub](#))

Deep Learning, Ensemble Methods, Computer Vision

- Built an ensemble of neural networks for road sign recognition, combining multiple models to improve classification robustness.

LLM-Powered Browser Automation Agent

TypeScript, Playwright, LLM APIs, REST APIs

- Built a production-ready browser automation agent pairing an LLM reasoning loop with Playwright to autonomously extract and act on data from evolving web pages, completing multi-step forms without manual intervention.
- Built a typed action-execution layer (click, scroll, fill, dropdown selection) with input validation and edge-case handling, exposed via a REST API supporting scheduled, unattended execution.

Movie Recommender System

Scikit-learn, NLTK, Collaborative Filtering

- Developed a hybrid recommendation pipeline combining collaborative filtering, content-based features, and Pearson correlation on IMDB metadata, with preprocessing for missing values and unstructured text.

OpenClaw Telegram Bot

OpenClaw, Telegram Bot API, LLM Agent Orchestration

- Built a personal Telegram bot assistant using the OpenClaw agent framework in a hands-on workshop, applying LLM-based conversational orchestration techniques.

PUBLICATIONS

Your Recourse, My Loss? Algorithmic Recourse under Shared Constraints

Zahra Khotanlou, Kate Larson, Amir-Hossein Karimi — [ACM FAccT 2026, Full Paper](#) (also [NeurIPS Workshop on Algorithmic Collective Action](#)).

RecourseBench: Reproducible Counterfactual Explanation Evaluation ([arXiv](#))

Zahra Khotanlou, Hashir Ahmed, Chenghao Tan, Ahmed Abdelaal, Amir-Hossein Karimi — *Preprint, under review.*

ToxTrans: A Deep Learning Method to Predict Protein Toxicity

Zahra Khotanlou, Hamed Malek — *Under review. Supervisor: Dr. Hamed Malek.*